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Project Title:	Dynamic Response of Particle-Filled Polymeric Fluids in Flows with Complex Lagrangian Time Histories		
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	Particle-filled polymeric fluids, frequently encountered in processing composite materials into fibers or sheets, turbulent drag reduction, and hydraulic fracturing, undergo alternating periods of simple shear and extensional motion in a Lagrangian reference frame. The dynamic rheological response of a particle-polymer mixture is not a simple superposition of independent polymeric and particle stresses. Instead, there is significant coupling between the velocity disturbance produced by the particles, polymer stress, and particle orientation. There is a dearth of relevant experiments and theoretical/ numerical analyses exploring the dynamic rheology of these systems. The proposed project provides an opportunity to employ theoretical and numerical simulation methods uniquely suited to predict the particle-induced polymer stress in transient flows and validate these methods by comparison with transient extensional rheology experiments of particle-polymer mixtures subject to pre-shearing in the Physical Sciences Informatics (PSI) database.		

Task Description:	A filament stretching rheometer used in the Shear History Extensional Rheology Experiments (SHERE, SHERE II, SHERE-Reflight / SHERE-R) enabled the measurement of a particle-filled polymeric fluid's transient extensional flow response subject to variable pre-shearing rates. SHERE and SHERE-R provide complementary measurements for the same fluid without particles. The experiments use dilute particle suspensions in a well-characterized Boger fluid over a range of moderate extension rates, conditions unachievable in Earth's gravity. A statistical analysis of the SHERE data indicates a significant reduction in strain hardening by particles at high strains, which agrees with a preliminary theoretical analysis. Prediction of the coupled particle and polymer stresses is arduous because the particle-induced velocity disturbance greatly alters the polymer stress in regions close to the particle surface, leading to strong polymer stress gradients that are difficult to resolve numerically and subtle to predict theoretically. The interplay between particle orientation, velocity disturbance, and polymer stresses when fibers are suspended in polymeric fluids presents further challenges. The project will employ theory and numerical simulations developed in our group, uniquely suited to address these problems. A finite difference method will solve the inertia-less viscoelastic flow in a spheroidal coordinate system that conforms to and rotates with the surface of a spheroidal particle and applies a time-varying linear velocity boundary condition at a nearly spherical outer boundary far from the particle. A demonstration that the ensemble average of the linear perturbation to the polymer stress due to the particle velocity disturbance is zero is essential to obtain convergent bulk rheological properties from numerical simulations. A complementary theoretical approach will yield semi-analytical predictions of the suspension rheology valid in the limit of small polymer concentration, c. Comparison with the numerical simulation will reveal the range of c, over which this theory is quantitatively accurate, and provide useful qualitative predictions for larger c. The theory is less computationally intensive and avoids computational stability limitations typical of nonlinear viscoelastic flow computations at high Deborah numbers and polymer extensibilities. Hence, it facilitates exploring broader parameter regimes. These methods will be applied to study the rheology of (1) suspensions of spheres in transient extensional flow with and without preshearing for comparison with the SHERE experiments; (2) suspensions of spheroids experiencing a simple shear flow preceded and followed by a period of extensional flow to model the processing of fiber-filled polymeric films; and (3) suspensions of spheroids subject to simple shear flow alternating with short bursts of extensional motion as a model of turbulent drag reduction by the coupled effects of fiber and polymer additives.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Suspensions of particles within polymeric or viscoelastic fluids are integral to numerous manufacturing and industrial processes. Within these applications, these suspensions navigate intricate flows, ranging from shear caused by solid surfaces to extension through orifices. The purpose behind introducing particles, such as spheres or fibers, is often to confer specific attributes upon the final product, such as heightened mechanical strength or directional electrical conductivity. However, the interaction between particles and polymers within the fluid is a pivotal factor dictating the energy demands throughout the manufacturing process. This interaction significantly alters the rheological properties of the suspension in comparison to the particle-free polymeric fluid. This study provides quantitative predictions and mechanistic understanding of the effects of particle-polymer interactions on the flow resistance of such materials.
	The transient rheology of a dilute suspension of particles is important in many industrial applications, such as in the food industry, manufacture of consumer products and pharmaceuticals, and during fracking processes. Polymer solutions with embedded particles have been extensively studied using experiments, but there are few theoretical studies of such systems. Analytical studies of particles in a polymeric fluid involve the derivation of expressions for the rheology at small concentration or small Deborah number, De, (De - the product of polymer relaxation time and the shear / extension rate). Numerical simulations have been performed for spherical particles in shear flow, but these studies have shortcomings, especially in the computation of the particle-induced polymer stress (PIPS). There are only a few studies that perform simulations of spheroids in shear flow, and they only consider aspect ratios smaller than 10. Our numerical technique aims to obtain a more accurate value of the PIPS and study the rheology of high aspect ratio (10-25) spheroids. We have developed a numerical code to solve for the flow around a spherical / spheroidal particle in spheroidal coordinates using a finite difference method. The Stokes equation and polymer constitutive equation are solved in a body fitted coordinate system that rotates and translates along with the particle. Both the fluid and particle are inertia-free. We have added functionality to the code to study the rheology of a spherical / spheroidal particle in an external imposed linear flow. The rheology of a dilute suspension of particles in a viscoelastic fluid can be modeled by considering the flow around a single particle. For this case, the ensemble average stress needs to be computed to determine different rheological parameters such as the viscosity, and the first and second normal stress differences. The per-particle contribution to the ensemble averaged stress can be divided into two parts – the stresslet and the particle induced polymeric stress (PIPS). When computing the ensemble averaged stress in the suspension, the particle stresslet, which is a surface integral of solvent and polymer stresses at the particle surface, captures the stresses occurring within the particle. Interaction with the particle also changes polymer extension in the bulk electrolyte, resulting in the PIPS contribution to the ensemble averaged stress. Some previous studies do not consider the PIPS contribution, which can lead to errors while determining the per-particle viscosity. The PIPS is calculated by integrating the non-linear polymer stress over the entire domain, including the fluid and the particle. The non-linear polymer stress is computed by subtracting the undisturbed polymer stress (polymer stress in the absence of particles) and the linearized polymer stress from the total stress. The linearized polymer stress is obtained by linearizing the polymer constitutive equation about the undisturbed stress and the imposed velocity. Subtracting the linearized polymer stress is very important, as the integral would not converge otherwise. Instead, it would lead to a logarithmic divergence, which manifests as a dependence of the integral value on the radius of integration. This could be a cause for errors in the computation of PIPS in earlier studies. After subtracting both the undisturbed and the linearized polymer stresses, volume integration of the non-linear polymer stress will provide an accurate value of the ensemble averaged PIPS. We use the Giesekus model to determine polymer conformation in this study. The Giesekus parameter ‘alpha’ determines the polymer extension beyond which further stretching is inhibited. Computation of PIPS inside the particle needs additional treatment for high aspect ratio spheroids, in order for the solution to be numerically stable. Inside the particle, the velocity is 0 in the frame of the particle. The polymer constitutive equation at the center can be reduced to an ordinary differential equation (ODE),

Task Progress:	which can be solved as a function of time only, and not space. The solution of this ODE is used as a boundary condition at the innermost radial coordinate. In addition, the innermost radial coordinate is also increased slightly – from less than 1% of the radius of a sphere to 5-15% of the minor radius of a spheroid. We show that for a spheroid aligned with the vorticity axis in shear flow, this approach leads to a small 2% error. We build upon our previous research on the extensional rheology of dilute sphere solutions in viscoelastic fluids. Our findings reveal that, across a wide range of polymer concentrations, introducing a small volume fraction of particles results in a negative contribution to the suspension stress due to particle-polymer interactions. Notably, the rate of this stress reduction is quadratic with respect to polymer concentration, effectively counterbalancing the linear increase in the undisturbed polymeric stress. This particle effect is particularly pronounced at higher polymer relaxation times or imposed extension rates, where the undisturbed polymeric stress is also elevated. Consequently, incorporating particles can lead to reduced energy consumption in industrial processes. Next, we consider the rheology of spherical particles, as well as the rheology and orientation dynamics of spheroids in simple shear flow. As a validation of our computations, the PIPS and stresslet contributions to the relative intrinsic viscosity are shown to compare well with results in the literature. Modest differences in the PIPS contribution to the intrinsic viscosity at large time and Deborah number may be attributed to the fact that, in the earlier results, the linearized polymer stress was not subtracted from the total polymer stress. Previous studies have contradictory claims about the regions around a sphere that contribute to the PIPS. We show that the radius of the region of integration for the PIPS shear stress should be increased when the Deborah number increases, in order to obtain the correct value of the integral. A parametric study of the dependence of the rheology of a dilute suspension of spheres in a shear flow on the fluid properties including the concentration of dissolved polymer, Giesekus parameter alpha, and polymer relaxation time (Deborah number) was performed for suspensions of spheres and of prolate spheroids that log roll about their vorticity-aligned axes of symmetry. Considering spheroidal particles aligned with the vorticity axis provides a simple way to determine the effect of particle aspect ratio and fluid properties on rheology in a simple steady state. For high polymer concentrations, the intrinsic viscosity scaled with the polymer viscosity decreases on increasing the polymer concentration, driven mainly by a plateauing and then decrease of the PIPS contribution. Increasing the Giesekus parameter also results in a decrease of the scaled intrinsic viscosity due to a decrease in the PIPS contribution. Increasing the Deborah number leads to an increase in the scaled intrinsic viscosity, while increasing the aspect ratio of a spheroid leads to a decrease in the intrinsic viscosity. A larger proportion of the area of a higher aspect ratio spheroid has a unit normal near the flow-gradient plane. Polymers in such regions have much reduced contributions to the stresslet owing to the time periodic shear flow they experience as they revolve around the spheroid's major axis. We investigate the orientation dynamics and rheology of high aspect ratio spheroids under steady shear conditions. Our numerical simulations at high polymer concentrations reveal that increasing the Deborah number initially slows the particle's motion in the flow-gradient plane, i.e., in tumbling orbit. For intermediate Deborah numbers, the particle comes to a standstill, first moving closer to the flow direction and then shifting away as the Deborah number increases further. At higher Deborah numbers, the particle resumes tumbling, with the tumbling motion accelerating with additional increases in the Deborah number. This behavior is likely due to the interplay between the polymer relaxation time and the Jeffery time period. Among the various stresses acting on the particle, the polymer-induced pressure is the most significant. For a log-rolling spheroid (aligned with the vorticity axis), the angular velocity initially decreases with increasing Deborah number before rising again. The polymer-induced strain emerges as the dominant force influencing the particle's motion. The log-rolling and tumbling motions are intricately linked to more complex three-dimensional orbits between these subspaces that can lead to initial orientation independent attractors, including a limit cycle near the vorticity axis, a fixed point near with the flow direction, or one or two fixed points situated between the flow and vorticity axes near the flow-vorticity plane. Finally, the complex long duration transient rheology associated with the coupled orientational dynamics and polymer stress fields around a spheroid in a three-dimensional orientation trajectory was explored. At low Deborah number, the particles spiral toward the vorticity axis executing approximate Jeffery orbits. This leads to short peaks of suspension stress as the particle flips, followed by longer periods of lower stress when the particle is near the flow-vorticity plane. The peaks decrease in amplitude as the particle approaches the vorticity axis. While the intrinsic viscosity is dominated by the stresslet, the PIPS also exhibits a peak while the particle flips. The PIPS takes on a small negative value when the particle is aligned near the flow-vorticity plane, possibly as a result of the changes in the particle forcing of the fluid as it passes from the compressional to the extensional quadrant of the simple shear flow. We have now explored the rheology of both shear and extensional flows separately. In the final months of the grant period, we will explore the rheology of suspensions in mixed flows that involve alternating periods of shear and extension. An example of such a flow occurs in the SHERE experiments when a pre-shear is performed before stretching the suspension in an extensional viscosity capillary rheometer.
Bibliography Type:	Description: (Last Updated: 10/15/2025)
Articles in Peer-reviewed Journals	Sharma A, Koch DL. "Finite difference method in prolate spheroidal coordinates for freely suspended spheroidal particles in linear flows of viscous and viscoelastic fluids." J. Comput. Phys. 2023 Dec 15;495:112559. https://doi.org/10.1016/j.jcp.2023.112559 , Dec-2023
Articles in Peer-reviewed Journals	Sharma A, Koch DL. "Rotation of a fibre in simple shear flow of a dilute polymer solution." J. Fluid Mech. 2023 Dec;976:A9.2023. http://dx.doi.org/10.1017/jfm.2023.823 , Dec-2023